

FriendTimes: Offering optimized and seamless gaming experiences with Google Cloud

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ABOUT FRIENDTIMES



To improve the quality of games distributed overseas, FriendTimes utilizes the high-performance server and speedy global network of Google Cloud to provide glitchless and highly available gaming

About FriendTimes

Founded in 2010, FriendTimes is an internet company that merges cultural creativity and technology R&D in its businesses across online game development and global operation, brand extension, investment in cultural and creative industry and mobile social platforms. Having received excellent business performance and reputation in China through several products, FriendTimes enjoys good public praise in Taiwan, Hong Kong and Macau region, South Korea, Japan, Southeast Asia and North America with a total of more than 139 million users.

Google Cloud results

- Lowers operational costs by 20% through optimizing VM usage rate
- Greatly enhances operation performance and maintenance efficiency with high-performance VMs on Compute Engine for game deployment
- Supports fast, error-free global content delivery and low latency with Cloud Load Balancing

No direct cause cyberattacks after migration to Google Cloud

Industries: Gaming

Location: China

Cloud CDN (<https://cloud.google.com/cdn/>),

About MeshCloud

MeshCloud is a leading cloud service provider and Google Cloud partner based in Beijing, China, offering cloud solutions to Chinese companies operating businesses abroad. By partnering with public cloud platforms and mastering the latest cloud technologies, MeshCloud helps its enterprise customers enhance business agility and lower cloud infrastructure costs.

About MeshCloud

Google Cloud (<https://cloud.google.com/>)

Google Cloud Armor (<https://cloud.google.com/armor/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Compute Engine (<https://cloud.google.com/compute/>)

Cloud Load Balancing (<https://cloud.google.com/load-balancing/>)

Monitoring (<https://cloud.google.com/monitoring/>)

infrastructure costs.

and Cloud

CDN

- Enables
detailed and
efficient
system error
detection
through
Cloud
Monitoring

Role-playing games (RPGs) are one of the most popular mobile game genres around the world with their exciting virtual adventures and immersive gaming experiences. However, despite countless offerings on the market, most of the role-playing games are centered around male characters and battles, which limits the diversity of gameplay.

FriendTimes (<https://www.friendtimes.net/en/index.html>)

is dedicated to diversifying the RPG scene with games featuring female characters and ancient Chinese aesthetics. Founded in 2010, the Chinese game developer currently operates nine popular mobile games for more than 100 million users across China, Northeast Asia, Southeast Asia and North America. Fate of the Empress

(<https://play.google.com/store/apps/details?id=com.GF.palacem5uscn.hwyad.google&hl=en&gl=US>)

, one of its flagship games, creates an intriguing ancient Chinese palace setting where gamers can customize their characters with various facial features and makeup styles, and explore different ways of living throughout the fascinating storyline.

"Since the foundation of our company, we've been focusing on developing mobile games that highlight female characters, which are rare on the market," says Hao Xiong, operations director at FriendTimes. "Through our appealing games, we want to raise the awareness of women's autonomy in the society and empower female gamers."

In 2012, FriendTimes started releasing its games outside China. At first, the company partnered with local operators and deployed its games on on-prem servers. However, as time went by, the performance of its on-prem server machines proved inadequate, which led to frequent downtimes requiring complex manual adjustments. Limited network bandwidth offered by the on-prem infrastructure also meant that they were at risk of DDoS attacks and unstable connectivity. To improve the gaming experiences that it offers, FriendTimes decided to move its games distributed overseas to [Google Cloud](https://cloud.google.com/) (https://cloud.google.com/) in 2021 for its high-performance virtual machines (VMs) and global network.

"To eliminate service disruptions and enhance maintenance efficiency, we needed to move our games distributed overseas to a cloud infrastructure with cost-effective computing resources and powerful global connectivity," explains Xiong. "Google Cloud checks all the boxes."

Reducing operational costs while eliminating maintenance downtimes

With technical advice provided by MeshCloud (<https://www.meshcloud.com/>) on system architecture, FriendTimes successfully migrated four games released in Taiwan and Singapore, as well as its account and update release services for overseas users, to Google Cloud. The company now uses VMs on Compute Engine (<https://cloud.google.com/compute>) for its overseas game servers. Since Google Cloud provides detailed information about the usage rate of each VM and suggestions on more cost-effective use, the FriendTimes team is able to lower 20% of infrastructure costs by optimizing the usage rate of its VMs.

The on-prem servers that FriendTimes previously deployed its games in Taiwan and Singapore on, used to experience system errors about two to three times every month, leading to downtimes that disrupted users' gameplay. On average, the company's operations team would take nearly an hour to repair errors, complete quality tests and resume normal operations. The high-performance VMs on Compute Engine have greatly enhanced the stability of FriendTimes' game servers. Even if an error occurs, the team can resume operations in a short time. After the migration to Google Cloud, FriendTimes has never encountered any technical issues with the VMs on Compute Engine, therefore improving the availability of its games and the user experience.

"Before, we deployed our games on on-prem servers and encountered downtime frequently due to hardware errors or decay. It required nearly one hour for us to resume our services, but users normally

couldn't wait for such a long time. On top of that, frequent downtime was detrimental to our gaming experience. The high-performance VMs of Compute Engine have helped us offer very reliable services by significantly enhancing the stability of our infrastructure and reducing the time required to clear errors," notes Xiong.

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Supporting faster and more stable connectivity with Cloud Load Balancing and Cloud CDN

Before, FriendTimes leveraged one physical server hosted in Mainland China to run its account and update release services for users outside China. When overseas users wanted to manage their accounts or update their game apps, the data needed to go through local networks to access FriendTimes' server, which often resulted in slow connectivity and connection errors, because the data transfer distance was long, and the network quality in different regions varied. When the connection was disrupted, users were not able to log in to the games or receive full update data.

Today, FriendTimes uses Cloud Load Balancing (<https://cloud.google.com/load-balancing>) and Cloud CDN (<https://cloud.google.com/cdn>) supported by Google Cloud's global network to offer account services and release game updates. As the data gets transferred via the intranet of Google Cloud, and Cloud Load Balancing and Cloud CDN rely on single anycast IP to deliver content from the server

that is nearest to end users, the content delivery speed has been significantly enhanced with zero data transfer error.

"With such a well-connected global network, Cloud Load Balancing and Cloud CDN have enabled us to guarantee fast and reliable content delivery. Before, the error rate of sending game configuration files was around 1.2%, whereas now our users can fully enjoy our games without disruptions caused by unsuccessful file delivery," says Xiong. "Overall, the network stability of our gaming services has increased from less than 99% to 99.99%."

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Preventing cyberattacks with Cloud Load Balancing and Cloud Armor

FriendTimes' games deployed in Taiwan used to frequently experience DDoS attacks, especially with a new game launch. Before the migration, the company's on-prem infrastructure was not able to mitigate or prevent DDoS attacks due to limited available bandwidth and the lack of security. In 2018, a game of FriendTimes newly released in Taiwan was down for three hours, greatly damaging its ability to attract users.

The game developer now employs Cloud Armor (<https://cloud.google.com/armor>) to detect DDoS attacks and filter out suspicious access requests, and Cloud Load Balancing to alleviate the impact of DDoS attacks by distributing traffic among servers. Previously, FriendTimes' game servers used to receive at least one malicious access request every day. After moving to Google Cloud, only a small number of the access requests sent by attackers could get through the company's servers, and no service disruption caused by DDoS attacks has ever occurred since.

Efficient infrastructure maintenance supported by Cloud Monitoring

To ensure high performance of its cloud infrastructure, FriendTimes uses Cloud Monitoring (<https://cloud.google.com/monitoring>) to detect errors and troubleshoot efficiently. Xiong says that Cloud

Monitoring allows the team to set up comprehensive alert mechanisms by providing detailed system operation information like incoming traffic sources and traffic distribution routes among game servers. This way, FriendTimes' engineers only need to check on the system when they receive alerts and are able to quickly clear errors.

"Before using Cloud Monitoring, we needed one to two engineers to constantly monitor our self-built alert system, and we're sometimes too late to react when an error occurs, because we didn't have all the system operation data," notes Xiong. "Cloud Monitoring has helped us maintain our services in a more efficient and effective manner."

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Unlocking the value of data to deliver better gaming services

Moving forward, FriendTimes plans to move its self-built game databases to Cloud SQL (<https://cloud.google.com/sql>), which can be seamlessly connected with BigQuery (<https://cloud.google.com/bigquery>), to analyze data related to ads and user behavior. As BigQuery requires lower costs compared with building a data analytics platform from scratch, the game developer will be able to conduct data analyses at a large scale in a more cost-efficient way to improve its gaming services. It will also deploy its new game that will be released in late 2022 in certain overseas regions on Google Cloud.

Xiong says, "With its high-performance cloud infrastructure, Google Cloud has helped us greatly enhance the stability and availability of our gaming services. We're looking to leverage more Google Cloud tools like data analytics, with the help of MeshCloud, to provide even better gaming experiences as our next step."